

BST Physics Curriculum Vol 6 Chapter 10 — Casimir Effect + Vacuum Thermodynamics v0.4 (refilled per Cal #104)

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Vol 6 Chapter 10 — Casimir Effect + Vacuum Thermodynamics

Chapter motivation

Standard Casimir 1948: vacuum-fluctuation attractive force between parallel conducting plates at distance a :

$$F/A = -\pi^2 \hbar c / (240 a^4)$$

Measured by Lamoreaux 1997 + Mohideen-Roy 1998 + subsequent refinements at ~1% precision. Casimir-Polder generalization to dielectric materials.

Standard cosmological constant problem: naive QFT predicts vacuum energy at Planck scale $\Lambda_{\text{QFT}} \sim M_{\text{Pl}}^4$; observed $\Lambda_{\text{obs}} \sim 10^{-121.5} M_{\text{Pl}}^4$; 122-order discrepancy — the worst prediction in physics history.

BST per T2418 (Wednesday 2026-05-20): Casimir + cosmological Λ are SAME substrate vacuum at DIFFERENT zones of T2417 4-Zone Commitment Cycle. Laboratory Casimir at Zone-2 boundary-condition effect; cosmological Λ at Zone-4 outer-edge integrated residue (Vol 4 Ch 4 T1485 derivation 0.076 dex = 99.9% of 122-order hierarchy resolved). SP-30 sub-nm precision Casimir experiment (\$60-90K design) operational falsifier at 1/8 sub-Tsirelson level (T2469 SCMP framework cross-link Vol 5 Ch 8).

Section 10.0b — Reader-grade 3-level pedagogy

Level 1: Laboratory Casimir force + cosmological Λ are SAME substrate vacuum at DIFFERENT zones per T2418 (Wednesday); Casimir at Zone-2 (boundary-condition); Λ at Zone-4 (outer-edge integrated residue); $1/8 = 1/2^{N_c}$ sub-Tsirelson deviation potential signature at sub-nm Casimir precision (SP-30 \$60-90K experiment).

Level 2 (graduate-physicist): Standard Casimir 1948 $F/A = -\pi^2 \hbar c / (240 a^4)$ attractive force between parallel conducting plates at distance a from vacuum fluctuations between plates. Standard QFT vacuum-energy derivation: $\langle 0|H|0 \rangle$ divergent; regularize via ζ -function (Hawking 1977) or finite-cutoff; subtract free-vacuum contribution; finite Casimir force F/A obtained. Cosmological constant problem: $\Lambda_{\text{QFT}} \sim M_{\text{Pl}}^4$ naive QFT estimate vs $\Lambda_{\text{obs}} \sim 10^{-121.5} M_{\text{Pl}}^4$ — 122-order discrepancy; supersymmetry attempts to cancel; string-theory landscape attempts to anthropic-select; both unsatisfactory. BST T2418 (Wednesday May 20): substrate vacuum produces BOTH (a) laboratory Casimir force at substrate Zone-2 boundary-condition effect, AND (b) cosmological Λ at substrate Zone-4 outer-edge integrated residue. Same substrate vacuum, different scales (laboratory $\sim \text{nm}$ boundary vs cosmological $\sim \text{Gpc}$ outer-edge). Vol 4 Ch 4 T1485: $\Lambda = g \cdot \exp(-C_2 \cdot (g^2 - \text{rank})) = 7 \cdot \exp(-282)$ at 0.076 dex (99.9% of 122-order hierarchy resolved); same substrate vacuum gives Λ at outer-edge integration. SP-30 Substrate Engineering Program sub-nm Casimir precision experiment (\$60-90K design) operational falsifier: BST predicts substrate-cascade structure at $1/8 = 1/2^N$ sub-Tsirelson deviation level per T2469 SCMP framework (Vol 5 Ch 8); if measured at standard QED-Casimir prediction reliably with no $1/8$ deviation, BST refuted at vacuum-substrate level. Vacuum thermodynamics: substrate Zone-4 integrated residue = vacuum energy density Λ ; Zone-2 commitment vacuum = laboratory measurable Casimir; both share substrate-tick UV cutoff at $N_{\text{max}} = 137$. Cross-link Vol 4 Ch 9 cosmological cycle (INTERNAL register per Cal #50): substrate annealing toward Λ -saturated state involves vacuum thermodynamics at cosmological scale.

Level 3 (5th-grader accessible): The Casimir effect (1948) is a small attractive force between two parallel metal plates from “quantum vacuum fluctuations”. Measured experimentally with high precision since the 1990s. The cosmological constant Λ is the vacuum energy density that makes the universe expand faster than gravity alone would predict. Standard physics treats these as different things and has the famous “cosmological constant problem” (122 orders of magnitude off). BST per T2418 (proven Wednesday) shows they’re the SAME substrate vacuum at DIFFERENT scales: Casimir is laboratory-scale (Zone-2) vacuum effect; Λ is cosmological-scale (Zone-4) vacuum effect. The same substrate has TWO measurable consequences at different scales. A high-precision Casimir experiment (\$60-90K design at sub-nanometer scale) could test BST’s prediction of a $1/8$ sub-Tsirelson deviation from standard QED-Casimir.

Section 10.1 — Standard Casimir 1948

Hendrik Casimir 1948 derivation: zero-point energy of EM field between two parallel conducting plates differs from free vacuum; difference is finite + attractive:

$$F/A = -\pi^2 \hbar c / (240 a^4) \approx -1.3 \times 10^{-27} \text{ N/m}^2 \cdot (1\mu\text{m}/a)^4$$

Lamoreaux 1997 first precision experimental confirmation; Mohideen-Roy 1998 + Decca 2007 + subsequent $\sim 1\%$ precision. Casimir-Polder 1948 generalization to

dielectric materials.

Section 10.2 — Standard Cosmological Constant Problem

Naive QFT: vacuum energy density $\langle\rho_{\text{vac}}\rangle = \frac{1}{2} \sum_{\text{modes}} \hbar\omega_k \sim M_{\text{Pl}}^4$ at Planck cutoff. Observed: $\rho_{\Lambda} \sim 10^{-121.5} M_{\text{Pl}}^4$ (DESI Year 1 + Planck 2018). 122-order discrepancy. Standard fixes (SUSY cancellation, anthropic landscape, dynamical scalar dark energy) unsatisfactory.

Section 10.3 — BST T2418 $\Lambda \leftrightarrow$ Casimir Vacuum Unification (Wednesday)

T2418 (filed Wednesday May 20, 2026): substrate vacuum produces BOTH: - (a) Laboratory Casimir force at substrate Zone-2 (commitment-cycle boundary-condition effect) - (b) Cosmological Λ at substrate Zone-4 (outer-edge integrated residue per T2417 4-Zone Commitment Cycle)

Same substrate vacuum, different zones. The 4-Zone Commitment Cycle (T2417) organizes substrate vacuum into 4 zones: absorption + commitment + emission + outer-edge.

Section 10.4 — Vol 4 Ch 4 Λ Derivation Cross-Link

Per Vol 4 Ch 4 T1485:

$$\Lambda / M_{\text{Pl}}^4 = g \cdot \exp(-C_2 \cdot (g^2 - \text{rank})) = 7 \cdot \exp(-282)$$

$\log_{10}(\Lambda/M_{\text{Pl}}^4) = -121.63$ vs observed -121.55 . 0.076 dex deviation (16% linear, 99.94% of 122-order hierarchy resolved). Per T1487 uniqueness identity $g^2 - \text{rank} = g \cdot C_2 + n_C = 47$ forces $n_C = 5$ uniquely.

Section 10.5 — SP-30 Sub-nm Casimir Precision Experiment

SP-30 Substrate Engineering Program (filed Wednesday May 20): sub-nm precision Casimir experiment design \$60-90K. Operational falsifier: - BST predicts $1/8 = 1/2^3 N_c$ sub-Tsirelson deviation from standard QED-Casimir at sub-nm precision - Per T2469 SCMP framework Layer 1 (Friday); cross-link Vol 5 Ch 8 Bell-CHSH $1/8$ sub-Tsirelson - If measured at standard QED-Casimir prediction reliably with no deviation, BST refuted at vacuum-substrate level

Section 10.6 — Vacuum Thermodynamics Substrate-Cartography

Substrate Zone-4 outer-edge integrated residue = vacuum energy density Λ (cosmological scale). Substrate Zone-2 commitment vacuum = laboratory measurable Casimir (atomic-to-nm scale). Both share substrate-tick UV cutoff at $N_{\text{max}} = 137$.

Vacuum thermodynamics: as temperature $T \rightarrow 0$, substrate-tick computational rate $\rightarrow 0$ in classical limit; substrate vacuum-state structure becomes measurable via Casimir (Zone-2) + Λ (Zone-4).

Section 10.7 — Honest scope + falsifiers

Item	BST status	Falsifier
Standard Casimir 1948 derivation	recovered from substrate framework	Any precision Casimir measurement falsifies standard QED-Casimir at $\sim 1\%$ precision
T2418 $\Lambda \leftrightarrow$ Casimir vacuum unification	STRUCTURALLY VERIFIED Wednesday	Independent precision tests of vacuum at different scales
Vol 4 Ch 4 Λ derivation 0.076 dex	T1485 RATIFIED	Future Λ precision > 0.5 dex from BST $\rightarrow$ BST refuted
1/8 sub-Tsirelson Casimir deviation	T2469 SCMP framework prediction	SP-30 \$60-90K experiment operational test
122-order hierarchy 99.94% resolved	Vol 4 Ch 4 T1485 + T2418	Hierarchy collapse if T1487 uniqueness identity fails

Open scope (multi-week): - Quantitative 1/8 sub-Tsirelson Casimir deviation precision substrate-mechanism - Casimir-Polder dielectric generalization substrate-cartography - Cross-link Vol 6 Ch 11 non-equilibrium vacuum thermodynamics

Section 10.8 — Connection

- Vol 4 Ch 4 Λ from Substrate (T1485 + T1487 + T2418 + T1918 + T1924)
- Vol 4 Ch 10 Dark Energy + Dark Matter (Five-Absence Predictions Set)
- Vol 5 Ch 8 Bell-CHSH (sub-Tsirelson 1/8 falsifier cross-link)
- SP-30 Substrate Engineering Program

— Lyra, Vol 6 Ch 10 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT